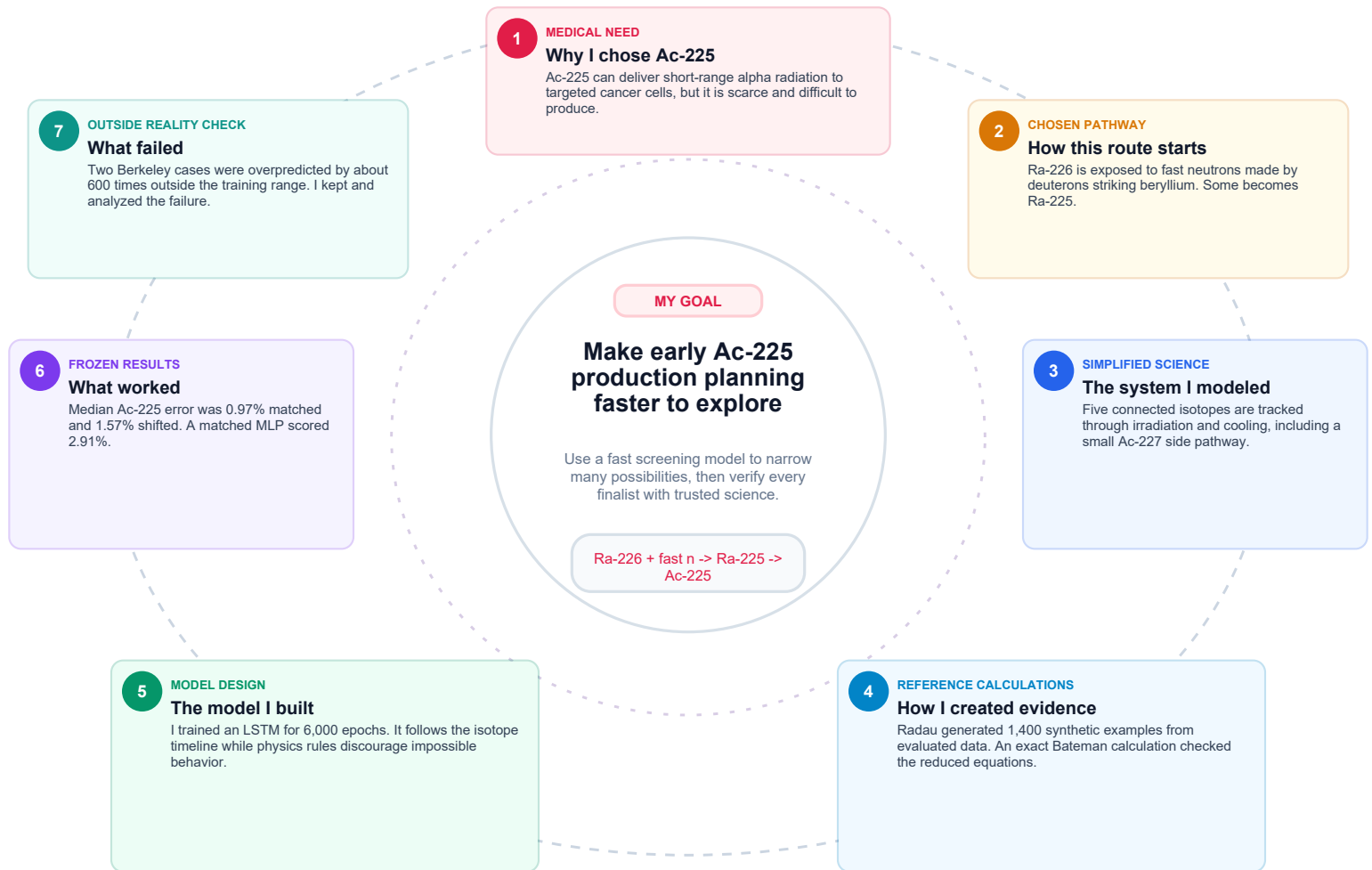


Can AI Help Screen Ac-225 Production Plans Faster?

I am testing whether a physics-guided computer model can narrow thousands of irradiation and cooling schedules before trusted nuclear calculations check every finalist.



WHAT I HAVE COMPLETED SO FAR

- 137-source scoping comparison
- Parameter-matched MLP comparison
- Reduced five-isotope physics model
- Untouched matched and shifted tests
- 1,400 Radau training examples
- 10,000-schedule decision audit
- V3 LSTM trained and frozen
- Berkeley outside stress test

WHAT I AM TRYING TO FINISH

- Define the exact planning decision.
- Obtain realistic spectrum, geometry, timing, recovery, and uncertainty data.
- Get expert review of the chain and reporting rules.
- Build a richer benchmark with an unseen final test.
- Present the successes and failures honestly.

Current evidence and limit: On 10,000 simplified scenarios, V3 recovered 18 of Radau's top 20 candidates and selected Radau's second-ranked plan, only 0.047% below the best score. It was 84.9 times faster than Radau but slower than the exact calculation for this small equation set. This is a screening prototype, not proof of real production savings or reactor accuracy.

Where AACPS support would help: Guidance on the science-fair pathway and documentation, help finding a qualified mentor to review the nuclear assumptions, and support connecting the project with appropriate university or laboratory resources. AACPS is not being asked to certify the model or provide radioactive materials.